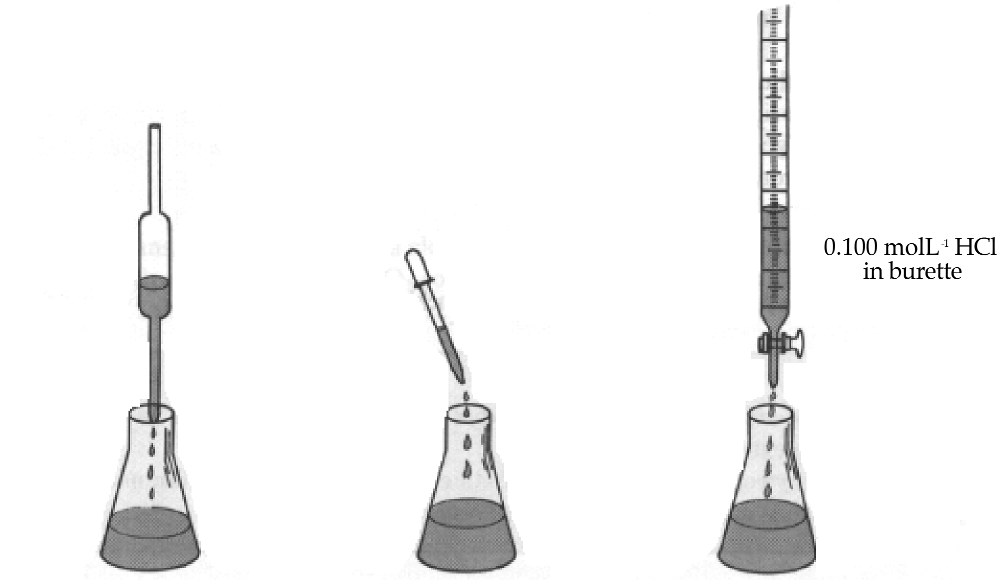
YEAR 10 BIOLOGICAL SCIENCE
BIOLOGICAL CHEMISTRY
TITRATIONS

AIM To determine the concentration of an unknown NaOH solution.

METHOD 1. Set up a burette with some 0.100 $molL^{-1}$ HCl solution.



Pipette out NaOH solution — Add phenolphthalein indicator — Titrate NaOH with HCl

2. Record the initial level of HCl accurately (to 2 decimal places).

3. Pipette 20.00 mL of the NaOH solution into a clean 250 mL conical flask and add 2-3 drops of phenolphthalein indicator.

4. Do a rough titration by running the HCl into the flask while swirling it. Turn off the acid when a colour change occurs.

5. Record the final level of HCl in the burette and determine the amount of acid added.

6. Refill the burette if necessary with HCl.

7. Prepare another 20.00 mL of NaOH solution and phenolphthalein in a clean conical flask.

8. Note the initial level of the HCl in the burette. Run the HCl into the flask to within 2.00 mL of the rough titration volume. Then add the acid drop-by-drop until a colour change occurs.

9. Record the final volume of the HCl in the burette, and then the volume of HCl added.

10. Repeat steps 7-9 for two other 20.00 mL samples of NaOH.

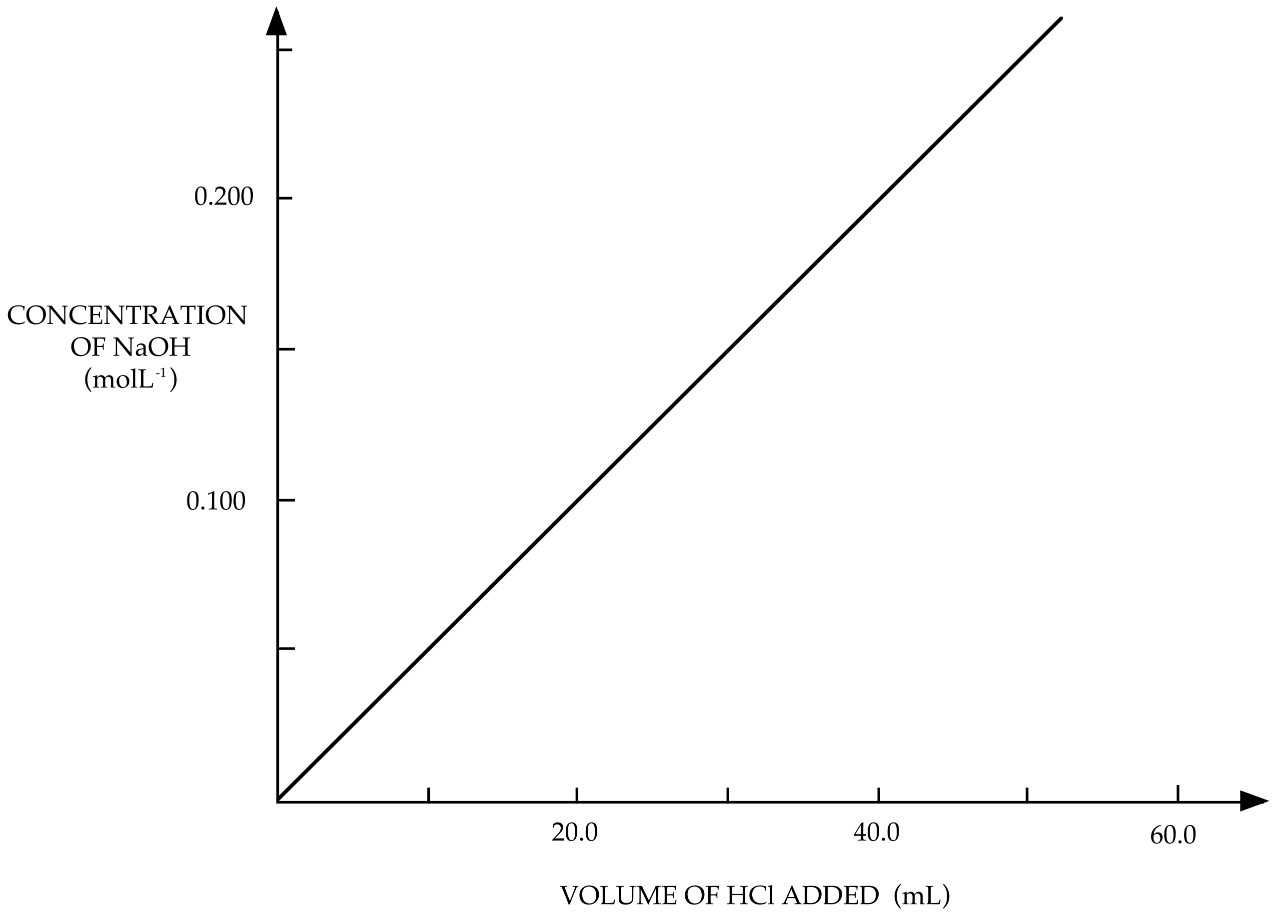
11. Average the accurate titration volumes to determine the amount of HCl required to neutralise the NaOH solution.

12. Use the graph supplied to calculate the concentration of the unknown NaOH solution.

RESULTS

VOLUME OF HCl ADDED (mL)	ROUGH ESTIMATE	ACCURATE TITRATIONS		
		1	2	3
FINAL READING				
INITIAL READING				
TITRATION VOLUME				

AVERAGE TITRATION VOLUME ________________



CONCLUSION

CRITICAL ANALYSIS